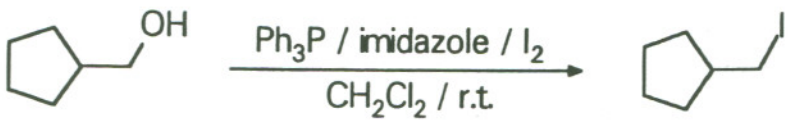
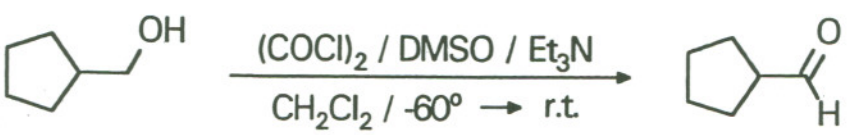
ORGANIC CUME
FEBRUARY 8, 2002

Propose a reasonable step-by-step mechanism for each of the following transformations. Show the structural formula for each important mechanistic intermediate. Be sure your mechanism explains the stereochemical result of the reaction if that is pertinent.

1.



2.



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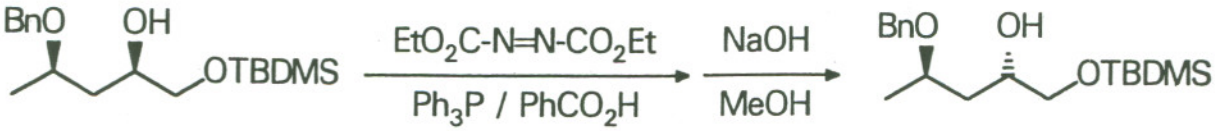
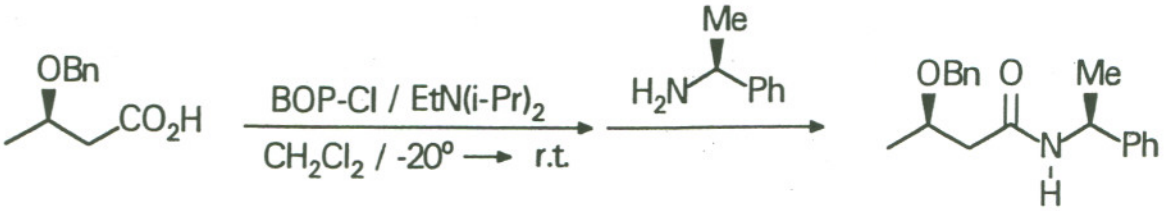
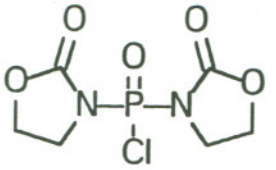


Bn = Benzyl; TBDMS = t-Butyldimethylsilyl

4.



BOP-Cl =



For answers see D. Seebach, M. A. Maestro, M. Sefkow, A. Neidlein, F. Sternfeld, G. Adam, and T. Sommerfeld, Helv. Chim. Acta, 74, 2112 (1991) and references therein.